# Runguo Li

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## Research Interests

**Machine learning systems for large models**: efficient LLM inference (sparse attention and KV-cache compression for long contexts, MoE serving), distributed training and RL post-training infrastructure, and the correctness of ML compilers and GPU kernels. More broadly: how systems constraints interact with model behavior, such as how memory budgets shape long chain-of-thought reasoning, informed by prior work on LLM reasoning, multimodal learning, and agent safety.

## Education

**University of Illinois Urbana-Champaign (UIUC)** — Aug 2026 – Present
*M.S. in Information Science (expected Jul 2028)* — Urbana-Champaign, IL, USA

**Shanghai University of Finance and Economics (SUFE)** — Sep 2022 – Jun 2026
*B.S. in Data Analytics, GPA: 3.78/4.0* — Shanghai, China

## Publications and Preprints

(* equal contribution)

[1] **Runguo Li**. **Certified Sparse Attention via Perturbed Retrieval**. *arXiv preprint arXiv:2610.24991, 2026.* [arXiv] [PDF]
Single-author. Sparse attention with per-head error certificates via perturbed retrieval; no certificate violated on three LLMs at 16K–128K tokens, with 4.6–7.9× lower 90th-percentile error than the best of 15 baseline families at equal keys read.

[2] **Runguo Li**. **Spend Bytes on Breadth: Precision–Count Trade-offs for Decode-Time KV Compression in Long Chain-of-Thought Reasoning**. *arXiv preprint arXiv:2610.05685, 2026.* [arXiv] [PDF]
Single-author. Splits a fixed KV byte budget between token count and precision (BreadthKV); beats eviction alone in 17 of 18 settings across three reasoning models and four benchmarks.

[3] Zhi Yang*, **Runguo Li***, Qiqi Qiang, Jiashun Wang, Fangqi Lou, Mengping Li, Dongpo Cheng, et al. **FinVault: Benchmarking Financial Agent Safety in Execution-Grounded Environments**. *arXiv preprint arXiv:2601.07853, 2026.* [arXiv] [Code]
First execution-grounded safety benchmark for financial agents (31 sandbox scenarios, 963 test cases); attack success rates reach 50%.

[4] Yeheng Chen*, **Runguo Li***, et al. **VeriBRT: Plan-Guided, Evidence-Based Automated Bug Reproduction**. *Under review, 2026.* [PDF]
Plan-guided bug-reproducing test generation; reproduces 341 of 433 issues on SWT-Bench Verified, 9.7 points above the strongest same-model baseline.

[5] Yutao Hou*, **Runguo Li***, et al. **Reasoning-Visual Critical Token Fine-Tuning for Multimodal Reasoning**. *Under review, 2026.* [PDF]
Selects which chain-of-thought tokens receive supervision using reasoning-relevance and visual-sensitivity signals (RVCFT).

## Research Experience

**University of Illinois Urbana-Champaign** — Aug 2026 – Present
*Research Intern, advised by Prof. Minjia Zhang* — Urbana-Champaign, IL, USA

- Efficient ML systems for large models: sparse attention with certified error for long-context decoding (single-author preprint [1]), decode-time KV-cache compression for long chain-of-thought reasoning (single-author preprint [2]), and kernel- and hardware-aware optimization.
- Upstream systems work grounded in this research: reproducing failures in production inference and training stacks and contributing fixes with regression tests (see Open-Source Contributions).

**Shanghai Jiao Tong University** — Mar 2026 – Jul 2026
*Research Intern, advised by Prof. Xiaodong Gu* — Shanghai, China

- LLMs for code: repository-level program understanding, automated bug reproduction, and coding-agent planning.
- Co-first author of **VeriBRT** [4], which represents issue-report behavior as a persistent Two-Axis Plan and uses reliability-aware evidence for bug-reproducing test generation; built the SWT-Bench Verified evaluation harness.

**Tencent Youtu AI Lab** Oct 2025 – May 2026
*Research Intern* Shanghai, China

- Co-first author of **RVCFT** [5], a selective-supervision method for multimodal chain-of-thought fine-tuning.
- LLM-driven adversarial data synthesis and low-resource, multi-label content-safety evaluation; contributed to a unified multilingual moderation model (XLM-RoBERTa backbone).

## Open-Source Contributions

22 merged pull requests across 14 ML-systems projects (project names link to the merged PRs):

*Inference and serving*

- **vLLM-Omni**: replaced the dense block mask in the Qwen2.5-Omni Token2Wav DiT with block-sparse attention, linear instead of quadratic in audio length; 3.5–6.4× lower per-request latency on H200 with matching outputs (#7975). Also turned an engine crash on a stray audio placeholder into a validation error (#7959).
- **vLLM**: fixed `int4_per_token_head` KV-cache quantization for non-power-of-two head sizes, where the Hadamard-rotation write aborted the engine core (#56198).
- **SGLang**: deferred DeepEP import-time failures to first use, so host misconfigurations no longer crash every server process, including dense models that never use DeepEP (#40671).
- **colibri**: multi-drive expert streaming for a 510 GB MoE checkpoint, physical-core OpenMP sizing (18.7× speedup in four engines), and a fix for 6.25 GiB of unbudgeted KV cache (#1517, #1522, #1526).

*Distributed training and post-training*

- **Megatron-LM**: routed MoE checkpoint shards with prepended layer/expert axes to the DCP checkpointable path instead of enumerating the global shard grid; 165× faster state-dict conversion and 3.4× shorter blocking phase of async save on 8×H200, with identical checkpoint metadata (#7287).
- **DeepSpeed**: fixed a ZeRO-3 rollout deadlock from unsynchronized generation stopping; blocked partial Hybrid Engine policy injection for unsupported architectures (#8264, #8265).

*Compilers and kernels*

- **PyTorch**: fixed an int32 overflow in the `cdist` CUDA backward kernel, where indices wrapped negative past $2^{31}$ and slipped past the bounds check into an illegal memory access (#198452).
- **FlashAttention**: fit the SM90 CuTe forward tile to the block-sparse block size; skipped the causal mask on unmasked KV blocks (580 → 477 inner-loop instructions on H200) (#2903, #2904).
- **Triton**: made nested-loop fusion stop trusting an `llvm.assume` outside the loop, which removed a zero-trip guard (#11521); lowered fp64 buffer `atomic_xchg` as an integer swap, which aborted compilation on AMD gfx942/gfx950/gfx1250 (#11904).
- **LLVM**: generalized ValueTracking to infer $x \neq y$ from $(x - y)$ compared with a constant (#225200); fixed an Attributor assertion from tracking non-pointer casts as pointers (#225290).

**Also**: JAX (batched `jax.scipy.linalg.lu`), Apache TVM (ONNX Hardmax with symbolic shapes), LMCache (allocator free-list corruption), and verl (FSDP critic loading).

## Honors and Awards

- National Encouragement Scholarship | People's Scholarship
- Shanghai Municipal Scholarship | Social Practice Scholarship

## Skills

- **Programming**: Python, C, CUDA, Triton, SQL, Bash.
- **Systems and frameworks**: PyTorch, JAX, vLLM, SGLang, DeepSpeed, Megatron-LM, FSDP, verl, MLIR, LLVM; GPU profiling on NVIDIA H200 and AMD MI250.